PyConDE
PyData
Darmstadt
2026

# From Pixel to Payouts: A Multi-Agent System for Real-Time Insurance Claims Processing

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**15/04/2026**

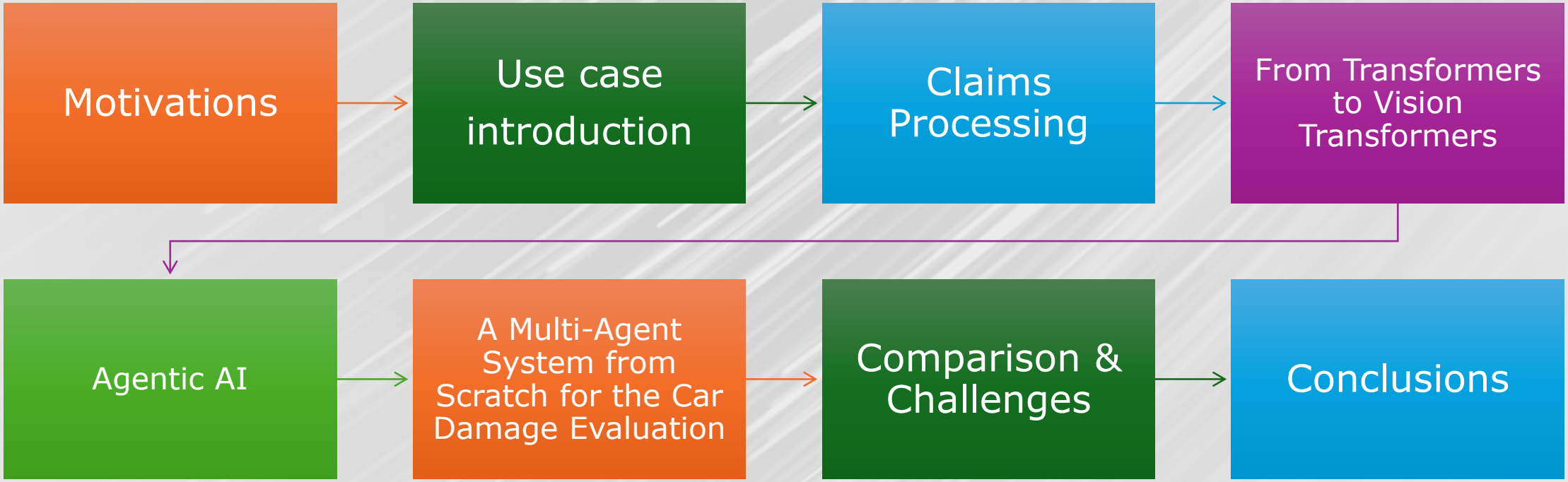
# MySelf:

Actuary during the day &
AI Scientist in the free time

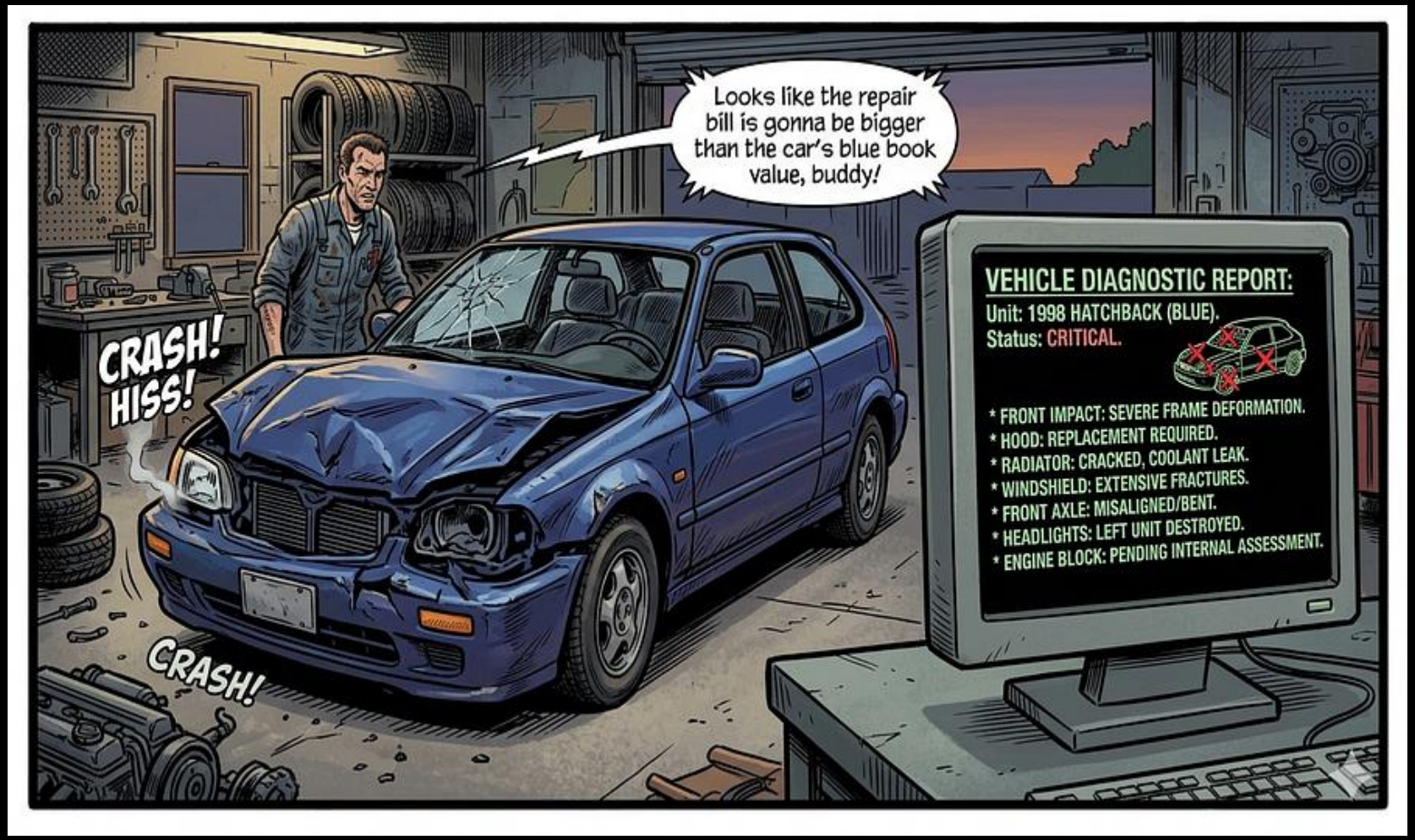
# Reach Me:

MyLinks

# Agenda

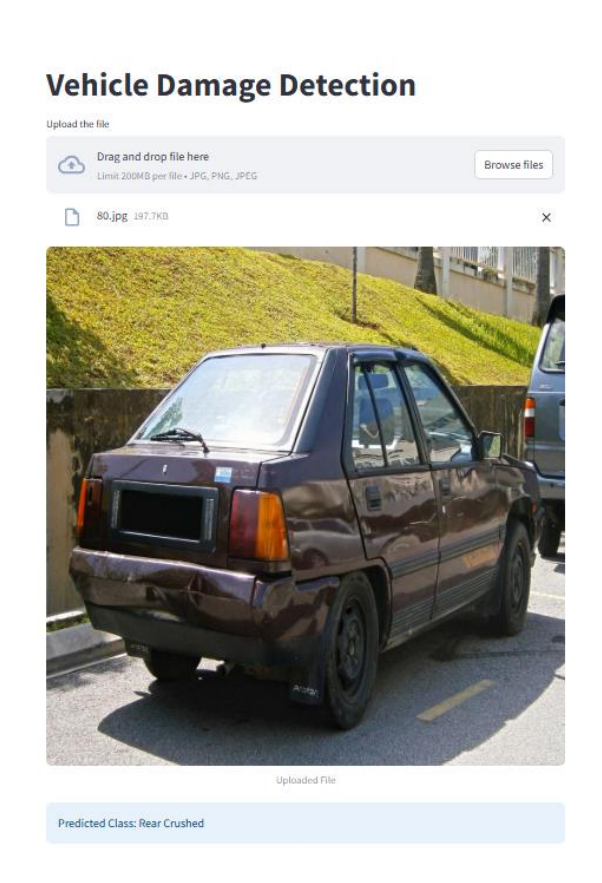

Motivations
Use case introduction
Claims Processing
From Transformers to Vision Transformers
Agentic AI
A Multi-Agent System from Scratch for the Car Damage Evaluation
Comparison & Challenges
Conclusions

# Motivations

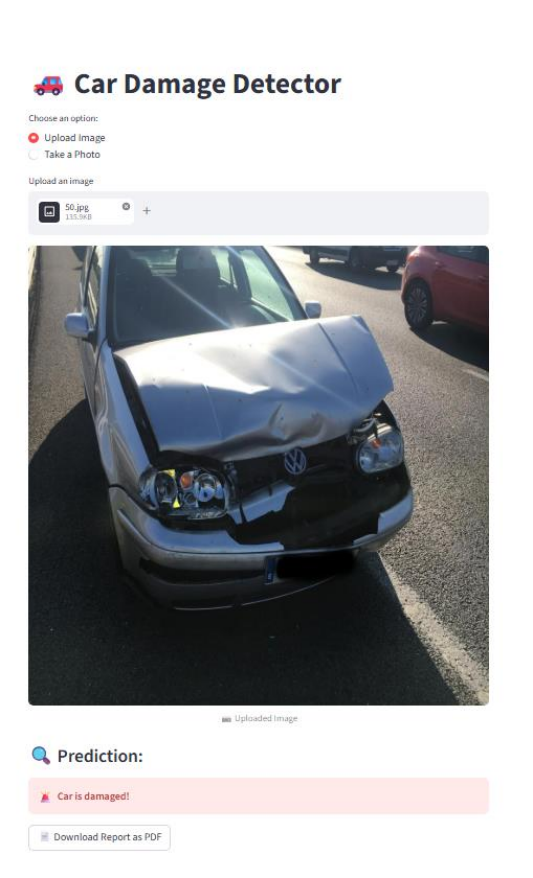



- Improve the old idea of a car damage classifier
- Employ a Multi-Agent System

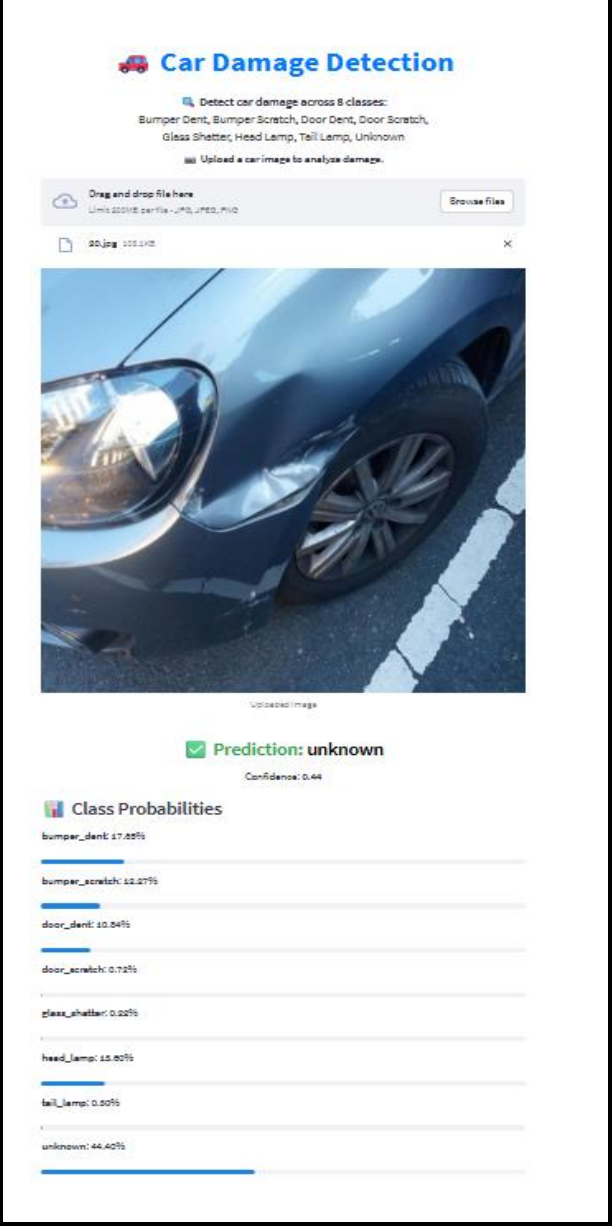
# The State of the Art around 2020




https://moksh-dl-car-damage-detection.streamlit.app

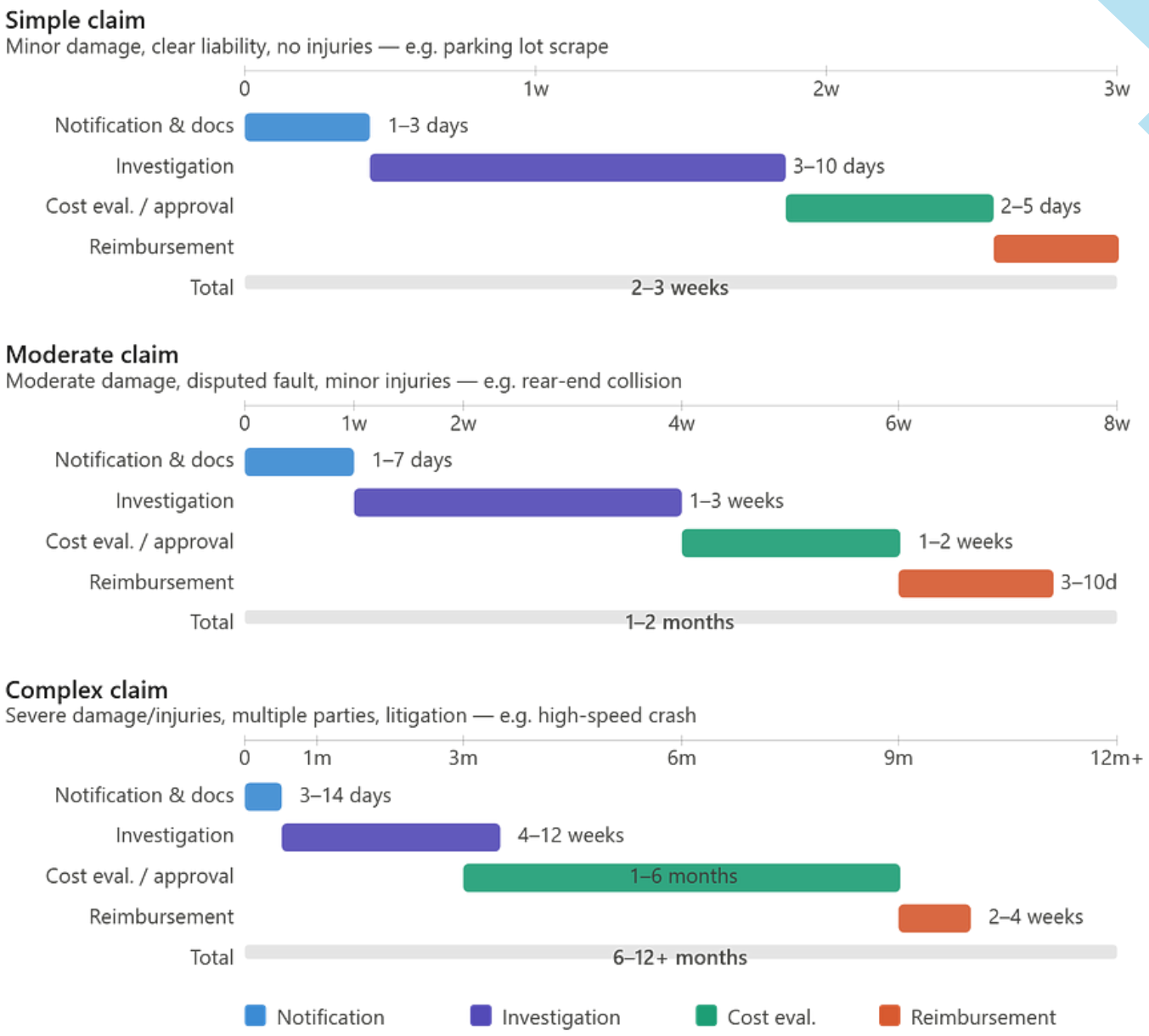



https://cardamagedetector.streamlit.app




https://car-damage-detect.streamlit.app

# Typical timeline of claims processing for a car accident



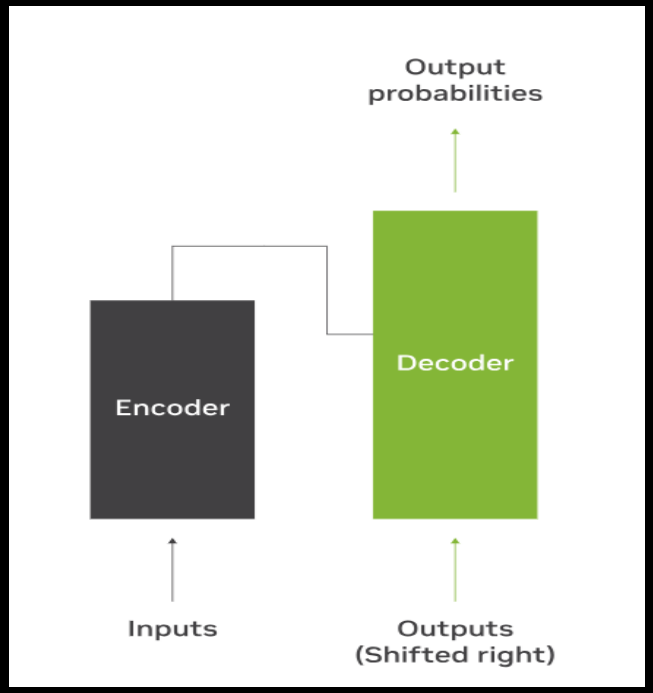

-Reference 1

-Reference 2

-Reference 3

# The Revolution coming from Transformers

The Transformer is a neural network architecture that uses self-attention mechanisms to process entire sequences of data simultaneously, enabling highly parallelizable and scalable training for tasks like language modeling and translation.




www.nvidia.com/en-eu/glossary/large-language-models/

- **Encoder-Only (Understanding)**

Example: BERT
**Use case:** classification, sentence similarity

- **Decoder-Only (Generation)**

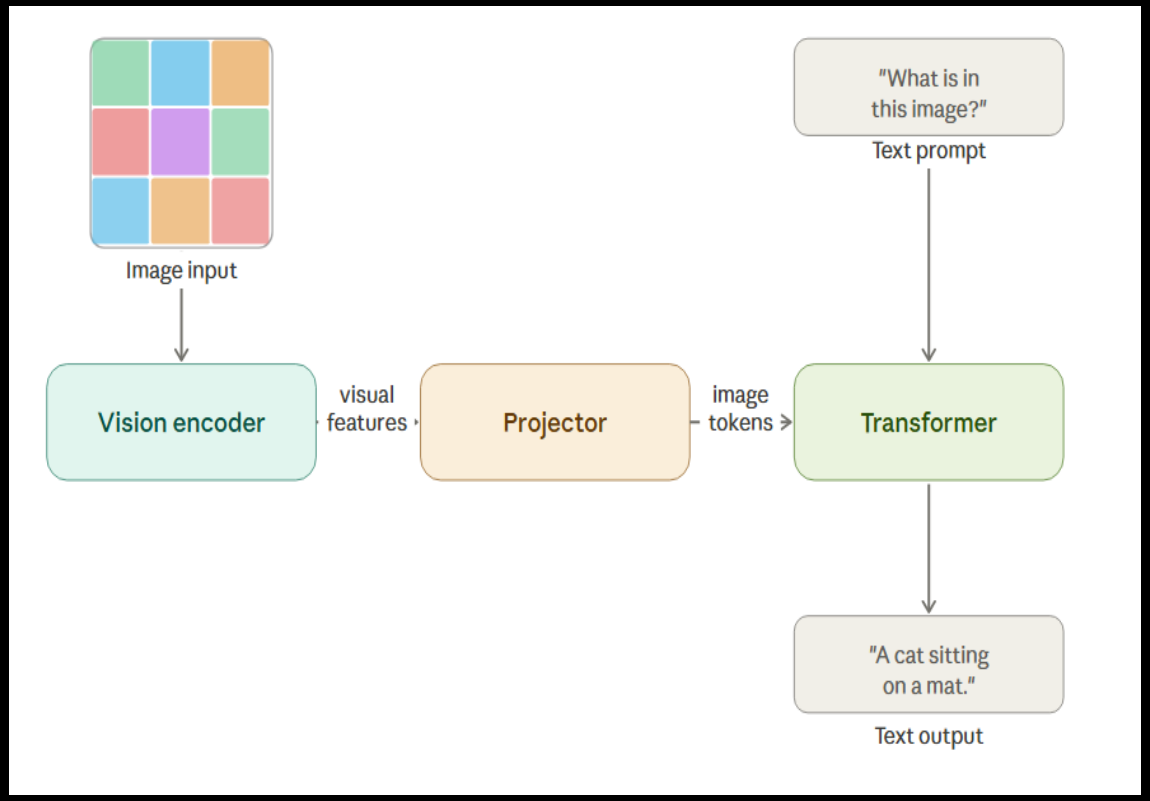
Example: GPT
**Use case:** text generation

- **Encoder–Decoder (Seq2Seq)**

Example: T5
**Use case:** translation, summarization

# The Revolution coming from Transformers

A Vision-Language Model (VLM) combines a Vision Encoder with a Transformer LLM, bridging image understanding and language generation to enable multimodal reasoning across text and images.

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- **Encoder-Only (Understanding)**

Example: CLIP
**Use case:** image-text retrieval, zero-shot classification

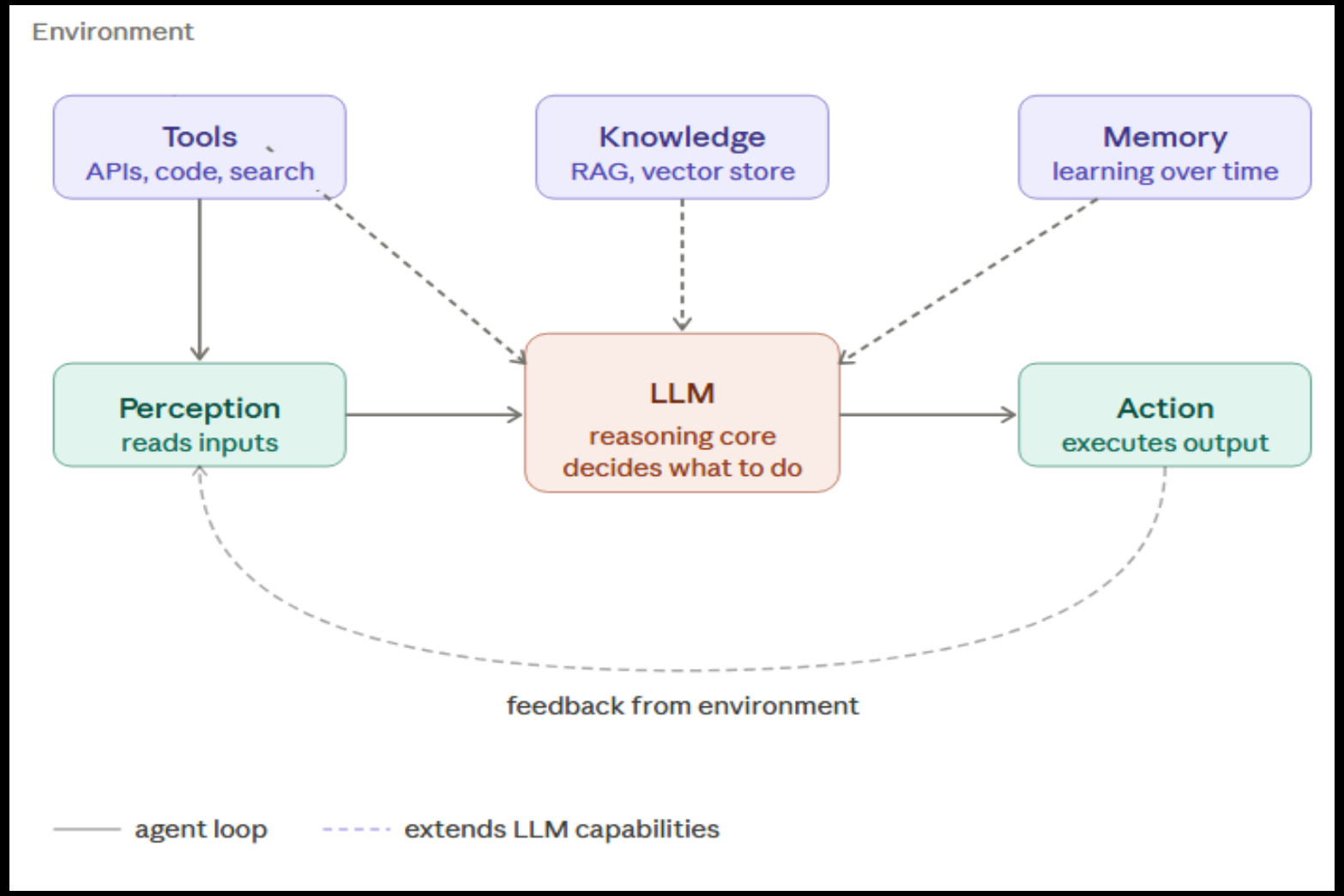
- **Decoder-Only (Generation)**

Example: GPT-5.2
**Use case:** image captioning, Visual Q&A

- **Encoder–Decoder (Seq2Seq)**

Example: Flamingo, PaLI
**Use case:** OCR, document understanding, translation

# From seeing to acting




A new era in which AI systems not only understand language but also reason, plan, and interact with their environment to perform complex tasks.

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Reasoning

Collaborating

# Why AI Agents?

**-> Autonomously Perform Tasks**
**-> Adapt and Learn**
**-> Increase Productivity and Efficiency**
**-> Enable Personalisation and Scalability**

Acting

Observing

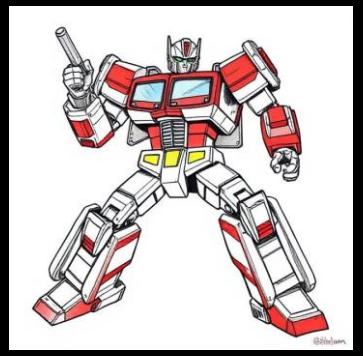
Self-Refining

Planning

# AI Agents Architecture

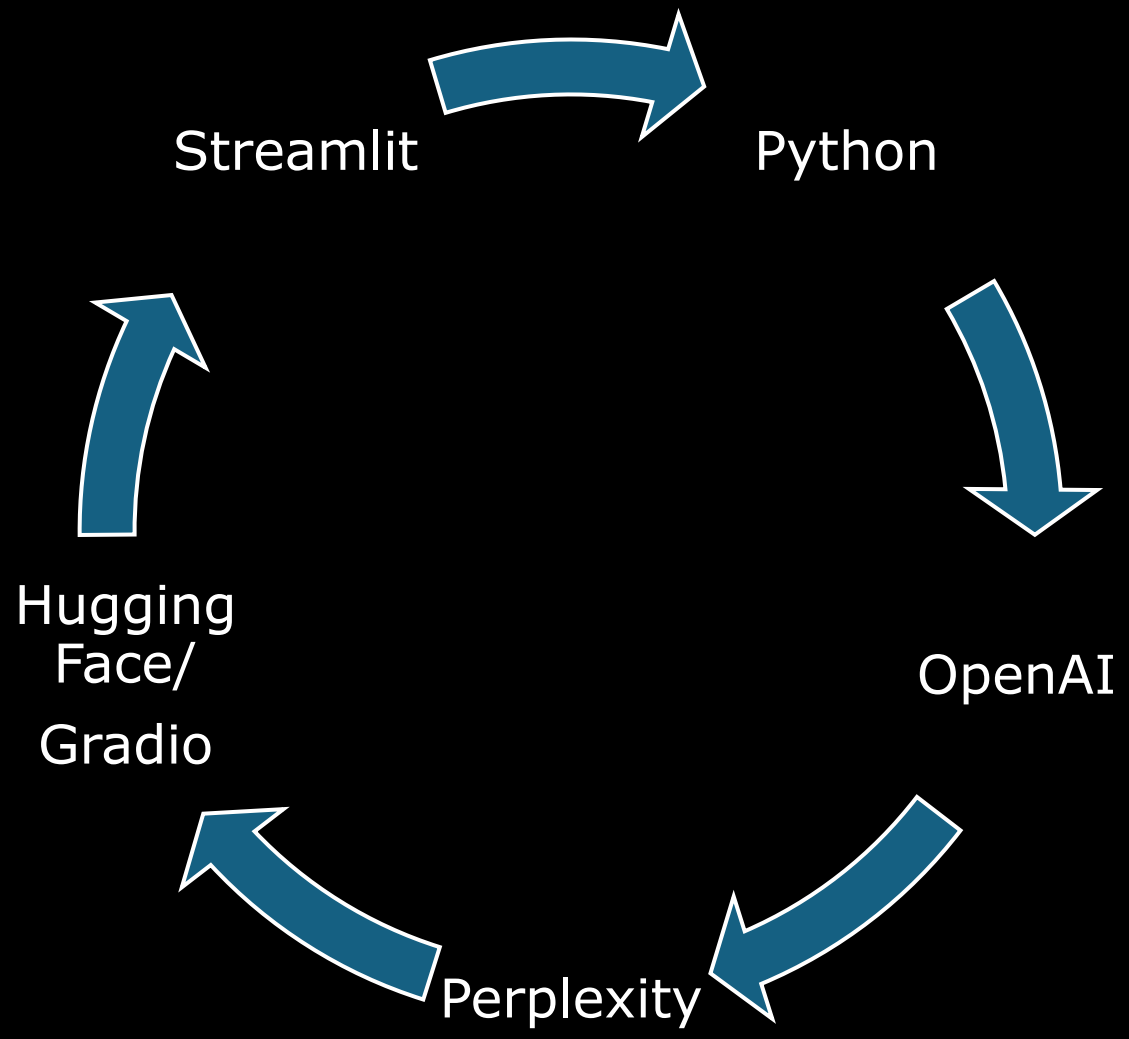
**Single-Agent**: One AI agent handles all tasks.
Simple to design and maintain, but less scalable
for complex tasks.

**Multi-Agent System (MAS)**: Multiple agents work together, enabling parallel processing and specialisation.
More scalable and robust, but introduces coordination challenges



https://deepai.org/machine-learning-model/text2img

# Tech Stack



# A Multi-Agent System from Scratch for the Car Damage Evaluation

```python
def analyze_damage_and_find_shops(perplexity_key, openai_key, image, location):
    # Step 1: Vision analysis
    damage_description = analyze_image_with_openai(openai_key, image)

    # Step 2: Cost estimation
    cost_data = estimate_cost_with_perplexity(
        perplexity_key,
        damage_description,
        location,
        currency
    )

    # Step 3: Shop search
    shops_text = find_shops_with_perplexity(perplexity_key, location)

    return description_output, cost_output, shops_output
```

At the end, a Multi-Agent System with a ReAct loop is achieved

```python
def react_run_loop(agent_state, thought_text, action_name,
action_fn, action_input=None):
    # Thought
    agent_state = react_thought(agent_state, thought_text)

    # Action
    agent_state = react_action(agent_state, action_name, action_input)

    # Execute
    result, observation_text = action_fn(action_input)

    # Observation
    agent_state = react_observation(agent_state, observation_text)

    return agent_state, result
```

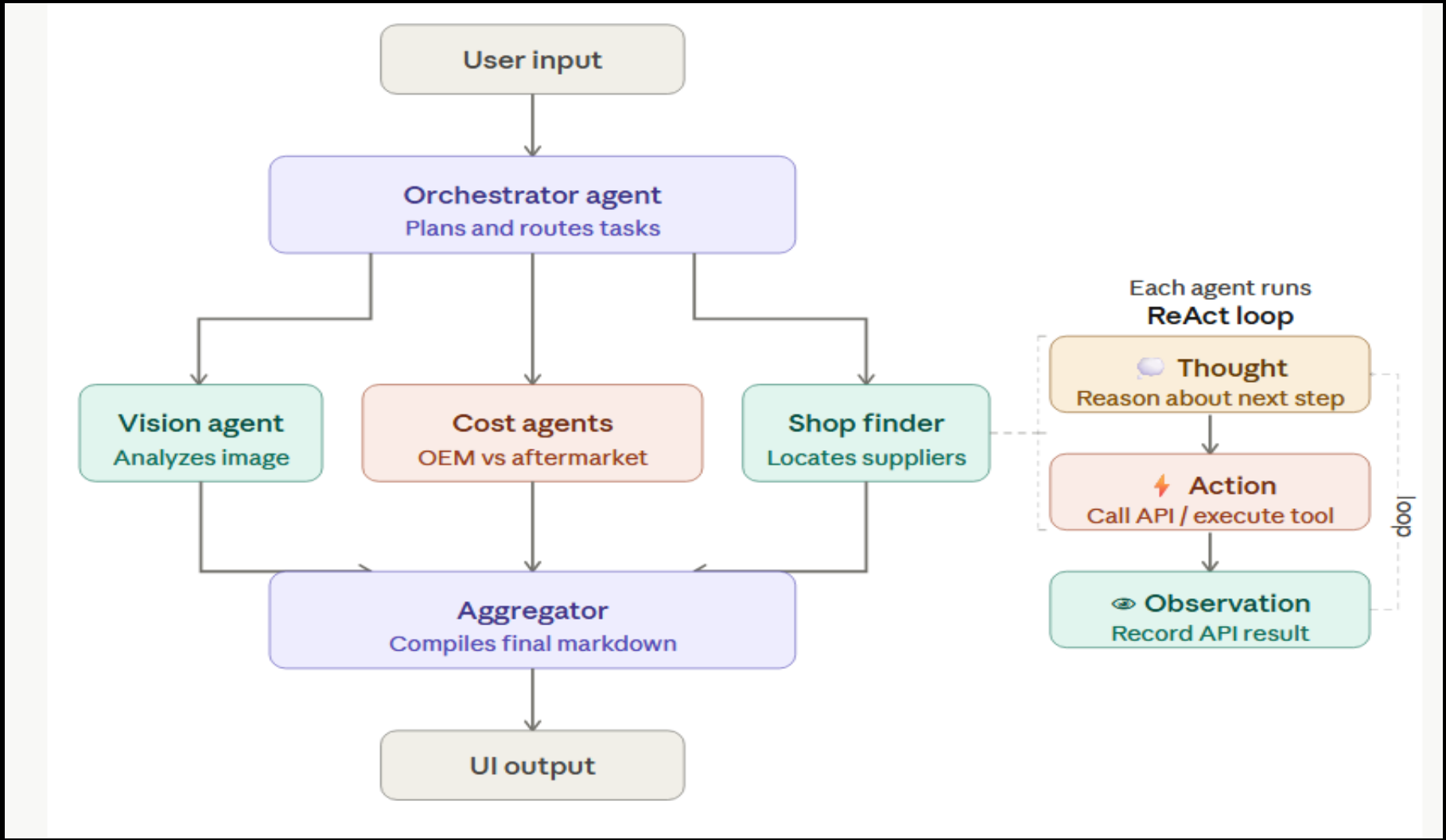
At the beginning, the project started with a simple APIs pipeline

# A Multi-Agent System from Scratch for the Car Damage Evaluation

```python
def create_agent_state(agent_type, config=None):
    """Create initial state for an agent"""
    return {
        'type': agent_type,        # What kind of agent this is
        'memory': [],              # Short-term memory: last 10 events
        'confidence': 1.0,        # Default confidence placeholder
        'last_action': None,       # Tracks what it did last
        'timestamp': datetime.now().isoformat(),# Creation time of the agent state
        'config': config or {},    # Agent-specific settings such as API key
        'performance_history': [], # Track record over time
        'react_trace': []        # Stores Thought → Action → Observation steps
    }
```

Each agent is stateful, explainable, and self-tracking, enabling autonomous and auditable decision-making.

# A Multi-Agent System from Scratch for the Car Damage Evaluation




Each agent keeps:

- Memory (last 10 events)
- Confidence score
- Configuration (API keys, settings)
- ReAct trace

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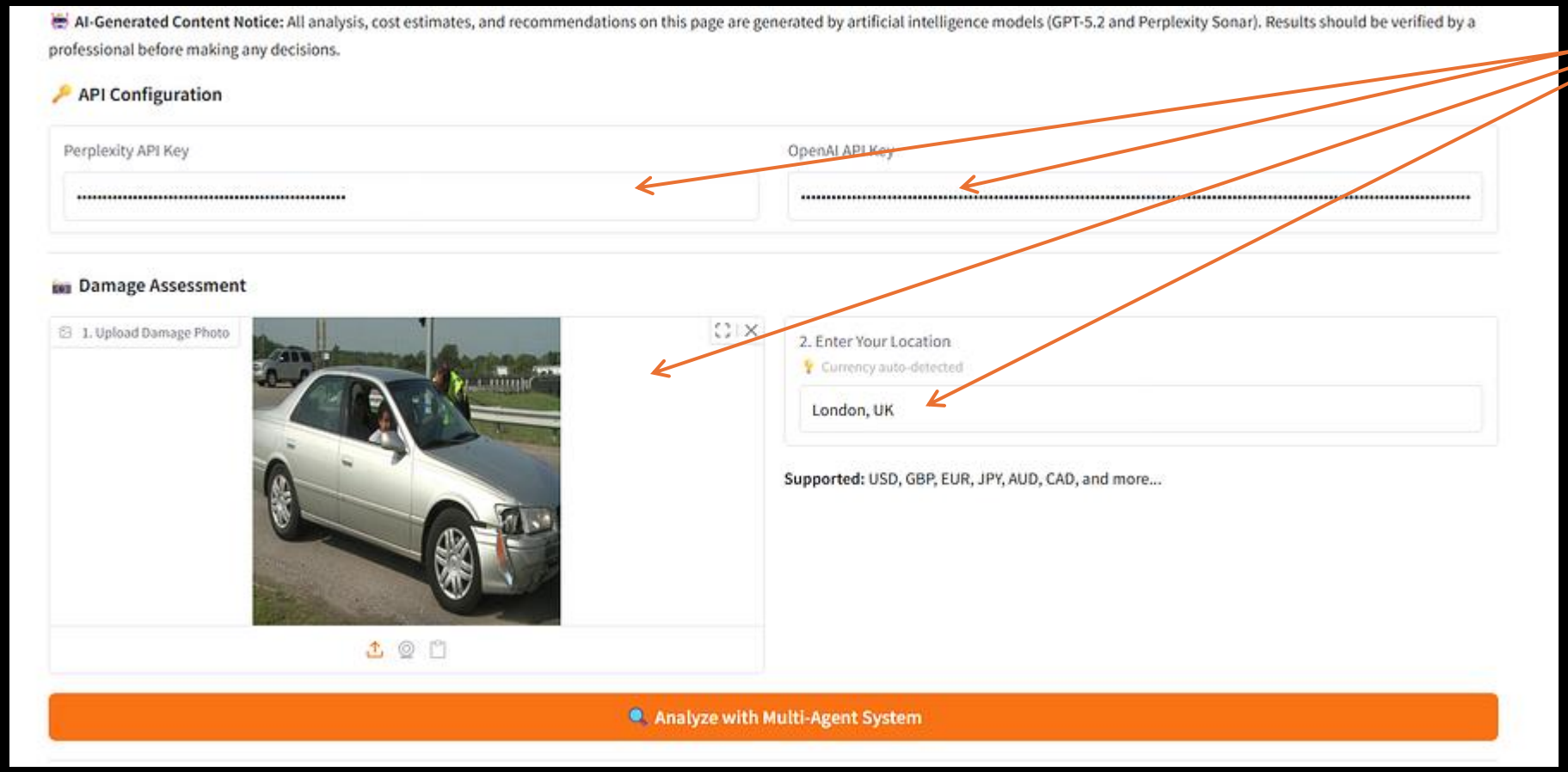
# A Multi-Agent System from Scratch for the Car Damage Evaluation




Input data

Go to the App

# A Multi-Agent System from Scratch for the Car Damage Evaluation

📊 Analysis Results 🤖 AI-GENERATED

🤖 AI-Generated Analysis

**Damage Analysis:**

Moderate front-right corner impact damage including broken right headlight assembly with missing lens and exposed components, torn front bumper cover separated from fender, displaced bumper retainer, deformed and scuffed right front fender at wheel-arch interface, and paint scuffs around damaged areas. No visible hood or windshield damage.

**Affected Parts:** Right headlight assembly, Front bumper cover, Right front fender, Bumper-to-fender retainers and clips, Paint and clear coat on affected areas

**Severity:** moderate

Damage Analysis from Vision Agent

💵 Comparative Cost Estimates 🤖 AI-GENERATED

🤖 AI-Generated Estimate

## 💎 OEM Parts - Dealer Service

**£1,800 - £3,200 GBP**

**Cost Breakdown:** At an Authorized Dealer in London, UK, using OEM parts, the repair begins with replacing the right headlight assembly, costing £800-£1200 for the genuine OEM unit including lens and inner components. The front bumper cover requires full OEM replacement at £600-£900 due to tearing and separation, plus £150-£250 for associated retainers, clips, and edge hardware. The right front fender needs repair or partial replacement with OEM panel at £500-£800, addressing deformation and scuffing at the wheel-arch. Labor at dealer rates of £180-£220 per hour totals 8-12 hours for disassembly, alignment, fitting, and reassembly, amounting to £1440-£2640. Painting and color-matching for the bumper, fender, and headlight surround adds £300-£500 using premium OEM-matched paint processes. Additional costs include alignment checks (£100-£200) and VAT at 20%, bringing the total to a premium-quality restoration with full manufacturer standards.

**Warranty:** Manufacturer warranty included

🤖 AI-Generated Estimate

## 💰 Aftermarket Parts - Independent Shop

**£850 - £1,650 GBP**

**Cost Breakdown:** At independent certified shops in London, UK, using quality certified aftermarket parts, the repair for this moderate front-right corner damage starts with a new aftermarket right headlight assembly costing £200-£350, as these are reliable non-OEM alternatives. The damaged front bumper cover right corner requires partial replacement or repair at £250-£450 including quality aftermarket part, plus bumper-to-fender retainers at £50-£100. Right front fender deformation and scuffing needs straightening, minor panel work, and refinishing, estimated at £200-£400 for the aftermarket repair materials and labor. Paint scuffs and matching around the bumper and fender edges involve sanding, priming, and respraying one partial panel plus blending, costing £150-£300. Labor totals 8-12 hours at London independent shop rates of £60-£80 per hour, covering disassembly, alignment checks, repairs, and reassembly, for £480-£960 overall. Additional costs include materials like fillers and paints at £50-£100, with no major structural work needed. Total reflects standard/good quality without OEM premiums.

**Warranty:** Shop warranty (typically 12 months)

Cost estimation from Cost Agents

# A Multi-Agent System from Scratch for the Car Damage Evaluation

AI-Generated Results

🛠️ **Local Repair Shops near London, UK**

Based on current search results, here are highly-rated auto body shops near London, UK specializing in collision repair:

## 1. Venom Auto Body Shop (London Auto Body Shop)

**Address:** 9 The Sidings

**Phone:** +44 20 8539 2741

**Rating:** Highly recommended[2]

**Business Hours:** Monday to Saturday: 8 AM – 8 PM; Sunday: Closed

**Services:** The team is noted for attentiveness and professional service in body repairs and collision work.[2]

## 2. JD Auto BodyWork

**Address:** 37 Padfield Rd

**Phone:** +44 7483 925095

**Rating:** Exceptional customer service[2]

**Business Hours:** Monday to Friday: 8:30 AM – 6:30 PM; Saturday: 9 AM – 2 PM; Sunday: Closed

**Services:** Scratch repairs, car resprays, dent removal, and comprehensive collision repair services. Customers praise their quick work and competitive pricing.[2]

Repair shops from Shop Finder Agent

# A Multi-Agent System from Scratch for the Car Damage Evaluation

🔍 ReAct Trace — Agent Reasoning Log

Shows each agent's Thought → Action → Observation loop. Expand after running an analysis.

🎯 **Orchestrator**

**Thought** 01:01:29

> I have received inputs with image=yes and location='London, UK'. I need to determine which agents to activate and in what order.

**Action** 01:01:29

> `build_execution_plan`

**Observation** 01:01:29

> Execution plan built with 4 steps: vision, cost_primary, cost_alternative, shop_finder.

**Thought** 01:01:29

> I have a plan with 4 steps. I will execute each agent in priority order: vision first, then cost estimators, then shop finder, passing results downstream as context.

**Action** 01:01:29

> `execute_agent_plan`

**Observation** 01:01:56

> All 4 agent tasks completed: vision, cost_primary, cost_alternative, shops.

ReAct log from the Orchestrator

# What about a comparison between Generative AI and Traditional Tools?

| Dimension | Traditional Tools (Rule-Based/CNN) | Multi-Agent Generative AI (MAS) |
|---|---|---|
| **Generalization** | Limited to inputs similar to training data; defaults to "nearest category." | Reasons about unseen damage patterns; produces novel, context-aware descriptions. |
| **Objective** | Single-function (e.g., "Is this a dent?"); answers one question at a time. | End-to-end assessment; simultaneously evaluates damage, severity, cost, and repair locations. |
| **Adaptability** | Static; requires manual updates or model retraining to reflect market changes. | Dynamic; retrieves live market data at query-time via search-powered agents (e.g., Perplexity). |
| **Explainability** | "Black Box" classification; produces an output without justification or logic path. | Transparent; provides natural language explanations and full **ReAct** (Thought → Action) traces. |
| **Modularity** | Monolithic scripts; the entire pipeline often needs retraining if one part changes. | Swappable modules; allows for a hybrid mix of GenAI and deterministic/proprietary tools. |

# Technical Constraints & Risk Management

| Risk Category | Constraint Detail | Mitigation / Requirement |
|---|---|---|
| **Financial Accuracy** | Estimates provided are **approximations** and AI-generated ranges. | Must be verified by a **qualified professional** before settlement. |
| **Input Dependency** | Vision accuracy is highly sensitive to **image quality** and resolution. | Low-quality photos require manual review to prevent downstream errors. |
| **Data Integrity** | Live retrieval (Perplexity) uses web sources, not a verified database. | Real-time pricing cannot be guaranteed as locally accurate without cross-referencing. |
| **Regulatory Compliance** | ReAct traces provide auditability but do not equal legal compliance. | Requires **formal human oversight** and documentation (e.g., EU AI Act alignment). |

# Conclusions

## What I've learned

- ✅ Building a Multi-Agent System from scratch is possible in Python without using a strict framework.
- ✅ A Multi-Agent System gives the opportunity to simplify complex dynamics in a valuable automation process for insurance claims.
- ✅ With a Multi-Agent System, you can employ different agents for specialized tasks and different specialized "brains".

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Generative-AI-in-Insurance-A-Game-Changer
GitHub Repository with links to the apps

# Thank you

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